only an answer that passes the test "survives" the decoherence. The failing answers cancel each other out "in a quantum way." Vague enough?

The series of qubits represents all possible solutions. This is simple enough to understand: a single qubit represents two possible solutions, and a hundred linked qubits mean 2^{100} possible solutions. The problem itself is formulated as a test to be applied to the potential answers. It is presented to the string of qubits so that they decohere, that is, "collapse" from their ambiguous states into real 1s and 0s that pass the test.

Now that's tough to digest, we know. We need an analogy. Ray Kurzweil, inventor, author, futurist, and strong-AI proponent, provides exactly what we need: when light strikes a mirror at an angle, it bounces off in the opposite direction at the same angle to the surface. But according to quantum theory, each photon actually bounces off *every possible point* on the mirror, thus trying out every possible path. The vast majority of these paths cancel each other out (in a quantum-interference way—now don't ask us how!), leaving only the "correct" path.

Now, in an abstract way, think of the mirror itself as the problem to be solved: "*what is it that makes the light bounce off at this angle?*" Only the correct solution—the light bounced off at the expected angle—survives all the quantum-interference cancellations. This is what we were talking about above—the *test of the correctness of the answer* is set up in such a way that most of the answers cancel out, and we're left with the one correct solution.

All At Once

You've probably gathered that quantum computing is all about massive parallelism—and we mean *massive*. We don't mean massive as in supercomputer-massive—we mean it as in giga-massive. A quantum computer doesn't just work "much faster" than a regular computer—it ropes in billions of our friendly neighbourhood universes to do in a matter of minutes what your rig would *literally* take aeons to do.

But then, that's not all good. It means we'll probably never have a general-purpose quantum computer—it'll only be good for tasks that *require* massive parallelism.

Decoherence Revisited

We've spoken about how decoherence is the process of removing the state ambiguity of a particle. But there is a more sinister face to the phenomenon: it prevents easy construction of a large-scale quantum computer.

So OK, how do we build one? We start with simple quantum logic gates, and integrate them into quantum circuits, of course! Easier said than done: gates and circuits are familiar enough at our level of size, but they're much more complex at the sizes where quantum effects come in. Still, we do already have quantum logic gates, simple devices that perform one elementary quantum operation, usually on two qubits. In fact, the first quantum logic gate was built using light in 2003. They differ from regular logic gates in that they perform operations on quantum superpositions (the ambiguous states of particles).

"The quantum theory of parallel universes... is not some optional interpretation emerging from arcane theoretical considerations. It is the only tenable explanation of a remarkable and counter-intuitive reality"

David Deutsch
Physicist
Oxford University

Now here are the practical problems: as we increase the size and complexity of the circuits we're merrily imagining, the number of interacting qubits increases, and this makes it harder to design the interactions that would exhibit quantum interference! (If you've been following, interference is *essential*.) One of the biggest impediments to progress is that *the surrounding environment* should not be affected by the interactions that give rise to quantum superpositions.

It's something like a process going on in a closed room; think of it as secrets being discussed inside. When the circuits get more complex, the "secrets" will spill to outside the room. Formally, this "spilling" of information to the outside is *also* referred to as decoherence. Now this is a totally different way of looking at the phenomenon. Don't ask us how the same thing can be thought of in two such different ways—that's the way it is!

To clear the fog for some readers, there is disagreement within the community—and different beliefs—about how to define decoherence. Different writers will tell you different things, and let's just leave it at this: if a particle can be in two states at once, why not a concept?

Breaking News

In October 2004, a quantum memory register using caesium atoms was built at the University of Bonn. Researchers were able to "write" to the register using microwave radiation, which puts the electrons into a position between their two natural orbits around the nucleus.

In July 2006, US NIST researchers brought quantum computers closer to mass production. The development reported was that of the construction of a two-dimensional ion trap; ion traps have proved to be the best way to make qubits, allowing up to eight to be connected together. (Kurzweil estimates that at 40 qubits, quantum computers will outdo todays PCs.)

In August 2006, a team of researchers at the Delft University of Technology in The Netherlands created a device that could manipulate a *single electron* using *conventional* chip fabrication technology. A major benefit of making a qubit this way is that the qubits will be easier to scale up.

So, then, we've got the memory, we've got the processors, we've got scalability. At this rate, we have good reason to believe we'll see a working quantum computer in our lifetimes!

Concluding Remarks

Quantum computing *is* the future. Repeat that a hundred times. But why? Well, for the simple reason that it's quantum mechanics that really governs the universe (or multiverse, if you will).

Our relatively feeble attempts at computing thus far have been limited to the application of rather elementary laws of physics. Even such things as DNA computing don't really exploit truly fundamental laws. After all, biology and chemistry *are* physics.

With quantum computing, we've reached the final frontier—the exploitation of the most successful theory of reality. We're admitting and harnessing the existence of parallel universes: we're taking hold of the very fabric of reality. ■

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